



QuantMesh x402 — Apne Dost Ko Pura Project Samjhao (Hinglish Explainer)

***Hey Teammate!** Is doc me apan ekdum easy, conversational Hinglish me **QuantMesh x402** ka full concept, features, inner working aur real-world use cases samajhte hain. Hackathon presentation ya teammate onboarding ke liye ye ekdum perfect guide hai!*



1. Bhai, Aakhir Ye Project Hai Kya? (Short Overview)

Dekh, simple words me: **QuantMesh x402** ek decentralized AI signal router hai jo Algorand blockchain par chalta hai.

Aajkal trading me signal lene ke do hi raste hote hain:

1. Ya toh \$100–\$300 har mahine subscription do (chahe mahine me ek trade lo ya zero).
2. Ya fir free Telegram/Twitter par fake/scam signals dekho.

Apan ne **Algorand ke x402 HTTP 402 (Payment Required) protocol** ko use karke **Pay-Per-Signal Model** banaya hai.

- User ko monthly subscription **NAHI** deni.
- Har fused AI signal ka cost sirf **\$0.007 USDC/ALGO** (approx ₹0.60 paise) hai!
- Aur sabse mast baat? **Agar hamara koi bhi AI bot (worker) down hua, toh user ka ₹1 bhi nahi katega! (Zero-Fee Guarantee).**



2. Humne Isse Kyun Banaya? (Problem Statement PS0404)

AlgoVerse 2026 Hackathon me **Problem Statement PS0404** tha: *x402 protocol ko AVM (Algorand Virtual Machine) par use karke pay-per-use DeFi intelligence product banao.*

Humne observe kiya ki baaki saari apps pehle paise kat-ti hain aur fir signal deti hain. Agar piche se API down hui ya model fail hua, toh user ka paisa gaya!

Humne **Pre-Execution Gating** lagaya:

*Orchestrator pehle 4 specialized AI bots chalta hai. ****Jab charo bots 100% success respond karte hain, TABHI server HTTP 402 payment challenge bhejta hai. Agar ek bhi bot fail hua, toh server sidha `HTTP 502` dega aur Lute Wallet prompt tak nahi hoga. Paisa zero charge hoga!*****



3. Key Features (Iske Main Highlights)

1. 🤖 4 Specialized AI Sub-Agents

Apan ne intelligence ko 4 alag microservices me baata hai:

- **Worker A (Sentiment Agent):** Financial news headlines ko HuggingFace ke **FinBERT Transformer NLP model** se pass karke sentiment score (0-100) nikalta hai.
- **Worker B (Whale Flow Agent):** CoinGecko Pro API se live volume, 24h/7d price change aur volume-to-market-cap ratio check karke **Whale Net Inflow/Outflow** nikalta hai.
- **Worker C (Technical Analysis Agent):** Live 1-day hourly candles fetch karke real math indicators calculate karta hai — **RSI, SMA 7/20 Golden/Death Cross, EMA 12/26**, aur **MACD**.
- **Worker D (Fusion Engine Agent):** In teeno ka mathematical weighted average karta hai $(\$30\% \text{ Sentiment} + 35\% \text{ Whale Flow} + 35\% \text{ TA})$ aur final **Composite Score (e.g. 66/100 BUY)** generate karta hai.

2. ⚡ x402 Micropayment Protocol

- HTTP 402 Payment Required status code use hota hai.
- Direct Lute Wallet se Algorand Testnet par \$0.007 atomic settlement hoti hai.

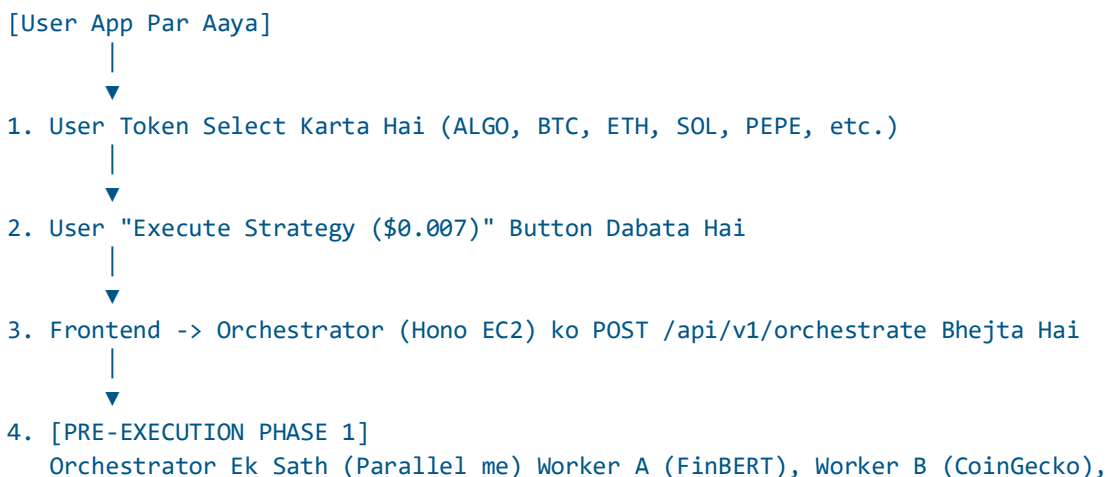
3. 📦 Algorand Box Storage Signal Attestation

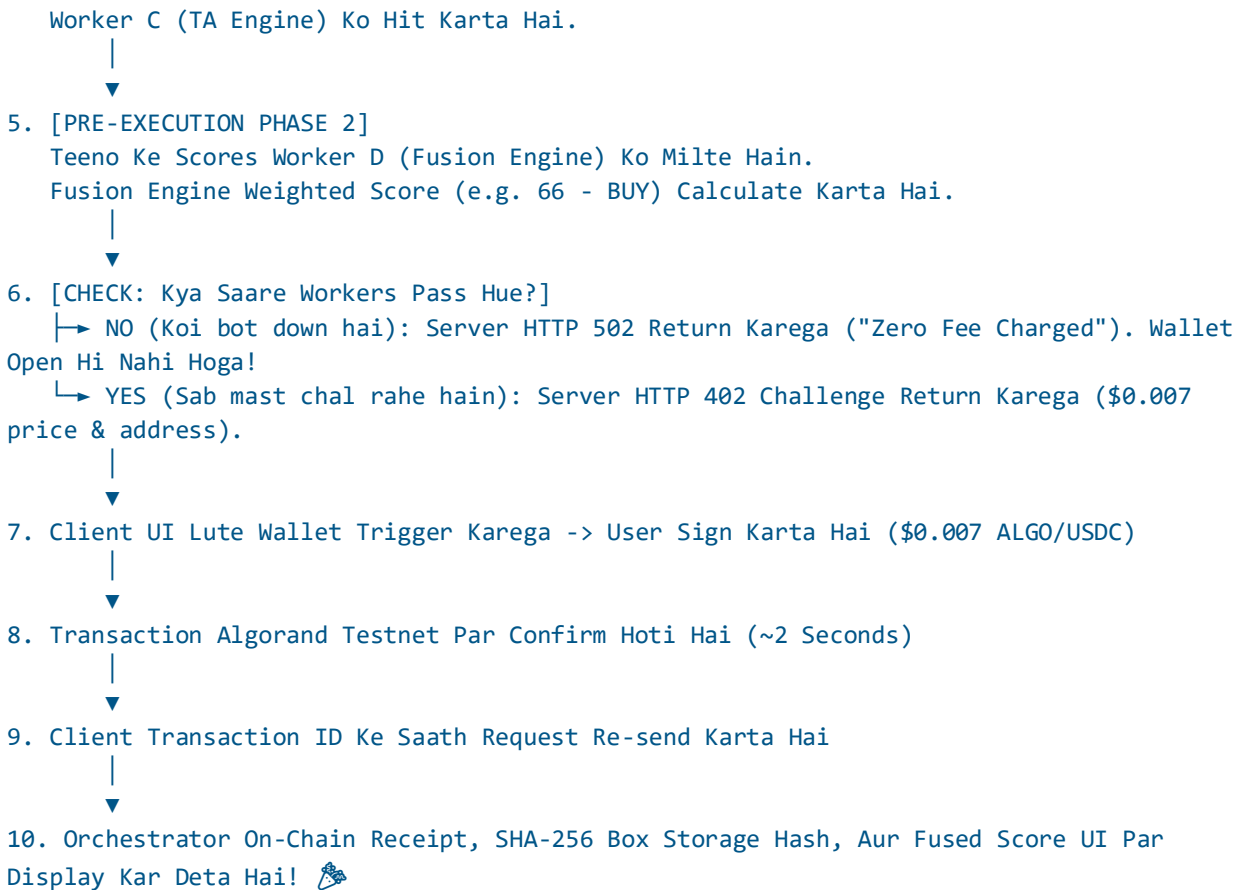
- Har generated signal ka ek unique 256-bit **SHA-256 cryptographic hash** banta hai (`sha256(token:score:verdict:txId:timestamp)`).
- Humne **PyTeal ARC-4 Smart Contract ('contracts/approval.teal')** banaya hai jo is hash ko Algorand Testnet ke **Box Storage** me permanently write kar deta hai. Pura audit trail on-chain rehta hai!

4. 📊 Dashboard Features

- **Animated Radial SVG Score Gauge:** Dynamic colors (Red = Sell, Yellow = Neutral, Cyan/Green = Buy).
- **Live Worker Health Heartbeat:** Har 30 seconds me `/api/v1/health` endpoint hit karke charo workers ka online/degraded status aur live latency (17ms-29ms) dikhata hai.
- **LocalStorage Signal History:** Purane saare signals browser me save rehte hain direct AlgoKit Lora Explorer link ke saath.
- **Interactive `/architecture` Page:** Pura flow visually samajhne ke liye dedicated route.

🔗 4. Pura System Kaise Kaam Karta Hai? (Step-by-Step Execution Flow)





5. Real-World Use Cases (Kaun Kaun Use Kar Sakta Hai?)

1. Retail Crypto Traders

- Jo traders mahine me 2-4 baar trade karte hain, unhe \$200/month subscription lene ki zaroorat nahi. Sirf jab signal chahiye, tab ₹0.60 paise do aur signal lo.

2. Algorithmic Trading Bots & Quants

- Automated trading bots (Hummingbot, custom Python scripts) humare x402 API endpoint ko hit karke programmatic micropayments ke zariye auto-trade execution triggers le sakte hain.

3. FinTech, MBA & Commerce Education

- MBA (Finance) & B.Com Students:** Financial NLP (FinBERT), Quantitative Technical Indicators (RSI, SMA, MACD), and Micro-SaaS Pricing Models ko live analyze karne ke liye ye ek perfect practical laboratory hai.

4. Quant Sub-Agent Marketplace (Future Expansion)

- Future me koi bhi 3rd-party developer apna custom AI sub-agent humare router par register kar sakta hai aur har signal execution par \$0.002 automatic revenue-share earn kar sakta hai!
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6. n8n Cloud Workflow Expansion Ideas (Visual WOW Factor for Demo)

n8n Cloud canvas presentation me bohot mast lagta hai! Extra technical depth dikhane ke liye n8n me ye naye sub-agents aur workflows add kiye ja sakte hain:

1. Macro Market Agent (`agent-macro`)

- **Kya karega:** Global macroeconomic metrics (Fed interest rates, DXY Dollar Index, S&P 500 correlation) pull karke macro market bias score (0-100) calculate karega.

2. Social Buzz & Velocity Agent (`agent-social`)

- **Kya karega:** n8n ke native Twitter/X ya Reddit nodes use karke selected token ki social mention volume spikes aur sentiment velocity (buzz) ko real-time track karega.

3. Algorand Ecosystem Liquidity Agent (`agent-liquidity`)

- **Kya karega:** Pact DEX ya Tinyman DEX ke liquidity pools se orderbook depth aur slippage query karke trade execution safety score nikalega.

4. Automated Telegram / Discord Alert Bot

- **Kya karega:** n8n workflow ke last me Telegram/Discord node add karke: jab bhi **"STRONG BUY"** signal (score > 75) generate hoga, n8n automatically ek stylish visual alert Telegram channel par post kar dega!

***Pitch Tip:** Presentation ke time side-by-side n8n canvas khol ke dikhao. Jab user website par "Execute Strategy" dabata hai, n8n canvas real-time me step-by-step green light up hota hai!*

7. Hackathon Pitching Ke Time Kya Bolna Hai? (Quick Summary)

3. **Tagline:** "QuantMesh x402 is an AI DeFi signal router powered by Algorand's x402 micropayment protocol."
4. **Main Innovation:** "Pay-per-signal at \$0.007 with a 100% Pre-Execution Zero-Fee Guarantee — if any AI sub-agent is down, the user pays zero."
5. **Live Tech Stack:** "Hono.js on AWS EC2, FinBERT Financial Sentiment NLP, CoinGecko Live Market Data, PyTeal ARC-4 Box Storage Smart Contracts, and Lute Wallet."
6. **Mainnet Ready:** "Only 2 config changes required to flip to Mainnet: setting network to `algorand:mainnet` and using Mainnet USDC ASA `315667040`."

QuantMesh x402 — Built for AlgoVerse 2026 Hackathon (PS0404).